



MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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Subject & Subject Code: Engineering Chemistry (1FY1-03)

Semester: I

Session: 2022-23 (Odd Sem.)

Branch: All

Batch: All

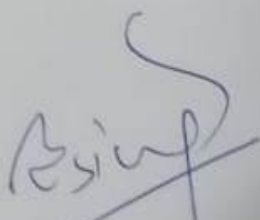
Unit No.: 3

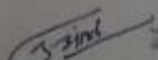
Date of Submission: -08.10.2022

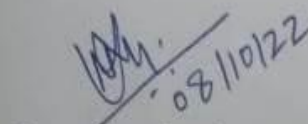
Lecture No.	Topics Covered	Plagiarism (%)	Unique (%)
1	Definition and significance of corrosion	33	67
2	Mechanism of chemical corrosion(dry)	68	32
3	Electrochemical (wet) corrosion,	65	35
4	Galvanic corrosion, concentration Corrosion and pitting corrosion.	48	52
5	Protection from corrosion; protective coatings- galvanization and tinning	59	41
6	Cathodic protection, sacrificial anode and modifications in design	39	61

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram


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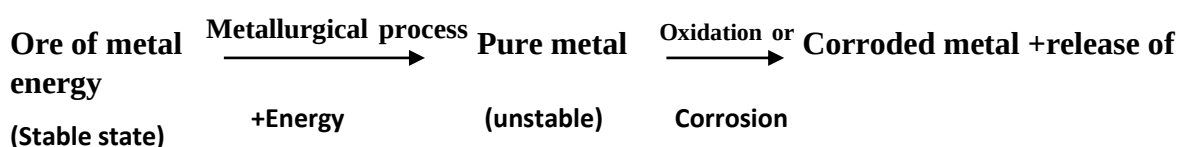
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UNIT-3 CORROSION AND ITS CONTROL

Corrosion - Corrosion of metals is defined as the spontaneous destruction of metals in the course of their chemical or electrochemical interactions with the environment. Thus, it is exactly the reverse of extraction of metals from ores.

OR

Corrosion is the process of gradual deterioration of metal from its surface due to unwanted chemical or electro chemical reaction of metal with its environment.



Example: Rusting of iron -A layer of reddish scale and powder of oxide (Fe_3O_4) is formed on the surface of iron metal.

A green film of basic carbonate [$\text{CuCO}_3 + \text{Cu}(\text{OH})_2$] is formed on the surface of copper, when it is exposed to moist-air containing carbon dioxide.

Although corroded metal is thermodynamically more stable than pure metal, but due to corrosion the metal losses its strength, ductility and finally it may fail suddenly under service condition.

CONSEQUENCES (EFFECTS) OF CORROSION:

The economic and social consequences of corrosion include

- i) Due to formation of corrosion product over the machinery, the efficiency of the machine gets failure leads to plant shut down.
- ii) The products contamination or loss of products due to corrosion.
- iii) The corroded equipment must be replaced.
- iv) Preventive maintenance like metallic coating or organic coating is required.
- v) Corrosion releases the toxic products.
- vi) Health issues (eg, from pollution due to a corrosion product or due to the escaping chemical from a corroded equipment).

CLASSIFICATION OR THEORIES OF CORROSION

Based on the environment, corrosion is classified into

- (i) Dry or chemical Corrosion
- (ii) Wet or electrochemical Corrosion

DRY OR CHEMICAL CORROSION:

This type of corrosion is due to the direct chemical attack of metal surfaces by the atmospheric gases such as oxygen, halogen, hydrogen sulphide, sulphur dioxide, nitrogen or anhydrous inorganic liquid, etc. The chemical corrosion is defined as the direct chemical attack of metals by the atmospheric gases present in the environment.

Example: (i) Silver materials undergo chemical corrosion by atmospheric H_2S gas. (ii) Iron metal undergo chemical corrosion by HCl gas.

TYPES OF DRY OR CHEMICAL CORROSION:

- 1. Corrosion by Oxygen or Oxidation corrosion
- 2. Corrosion by Other gases
- 3. Liquid Metal Corrosion

1. CORROSION BY OXYGEN/OXIDATION CORROSION:

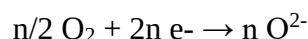
Oxidation Corrosion is brought about by the direct attack of oxygen at low or high temperature on metal surfaces in the absence of moisture. Alkali metals (Li, Na, K etc.,) and alkaline earth metals (Mg, Ca, Sr, etc.) are rapidly oxidized at low temperature. At high temperature, almost all metals (except Ag, Au and Pt) are oxidized. The reactions of oxidation corrosion are as follows:

Mechanism:

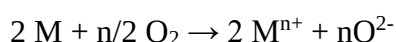
- 1) Oxidation takes place at the surface of the metal forming metal ions M^{n+}



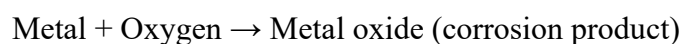
- 2) Oxygen is converted to oxide ion (O^{2-}) due to the transfer of electrons from metal.



- 3) The overall reaction is of oxide ion reacts with the metal ions to form metal oxide film.



The nature of the oxide formed plays an important part in oxidation corrosion process.



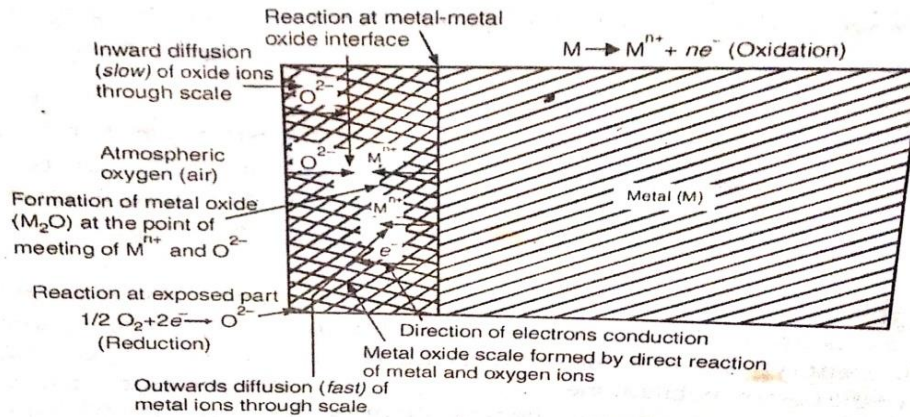


Fig. 1. Oxidation mechanism of metals.

When oxidation starts, a thin layer of oxide is formed on the metal surface and the nature of this film decides the further action. If the film is-

- (i) **Stable:** A Stable layer gets adhered tightly to the parent metal surface. Such a film behaves as protective coating in nature, thereby shielding the metal surface. The oxide films on Al, Sn, Pb, Cu, etc., are stable, tightly adhering and impervious in nature.
- (ii) **Unstable oxide layer:** This is formed on the surface of metals such as Ag, Au, Pt. As the metallic state is more stable than oxide, it decomposes back into the metal and oxygen. Hence, oxidation corrosion is not possible with noble metals.

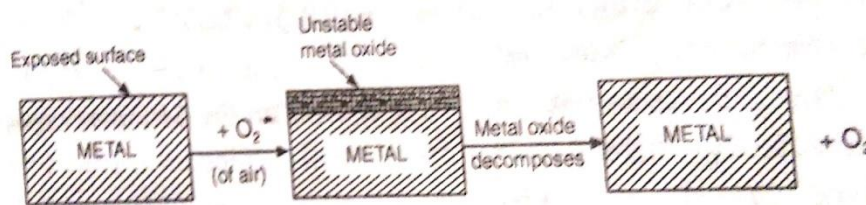


Fig. 2. Unstable oxide layer.

- (iii) **Volatile oxide layer:** The oxide layer film volatilizes as soon as it is formed. Hence, always a fresh metal surface is available for further attack. This causes continuous corrosion. MoO_3 is volatile in nature.

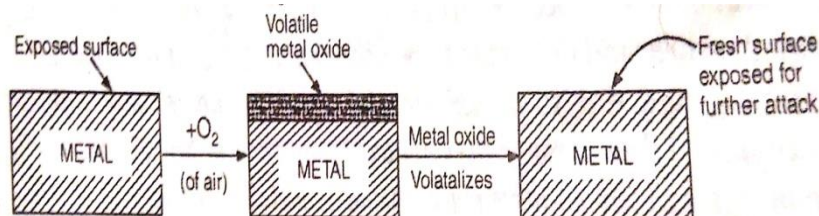


Fig. 3. Volatile oxide layer.

(iv) Porous layer: The layer having pores or cracks. In such a case, the atmospheric oxygen has access to the underlying surface of metal, through the pores or cracks of the layer, thereby the corrosion continues till the entire metal is completely converted into its oxide.

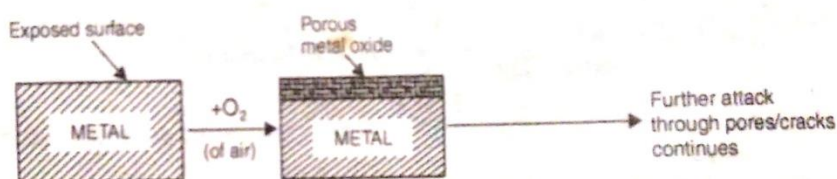


Fig. 4. Porous oxide layer.

Pilling – Bedworth rule: The oxide layer acts as a protective or non-porous barrier, if the volume of the oxide is more than the volume of the metal from which it is formed. On the other hand -if the volume of the oxide layer is less than the volume of metal, the oxide layer is porous and hence non-protective because it cannot prevent the access of oxygen to the fresh metal surface below.

$$\text{Specific volume ratio} = \frac{\text{Volume of Metal oxide}}{\text{Volume of metal}}$$

“Smaller the specific volume ratio, greater the oxidation corrosion”.

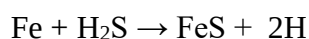
For eg: alkali and alkaline earth metals like Li, Na, K, Mg forms oxides of volume less than volume of metals. These layers are porous and non-protective. Hence these undergo corrosion more rapidly. On the other hand metals like Al forms oxide whose volume is greater than the volume of the metal. Therefore Al forms a tightly – adhering non-porous protective layer.

2. Corrosion by other gases – This occurs when dry gases like H_2 , Cl_2 , F_2 , CO_2 etc directly attack on surface of the metals or alloys.

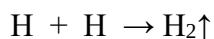
Example- By hydrogen

Hydrogen Embrittlement: Loss in ductility of a material in the presence of hydrogen is known as hydrogen embrittlement. This type of corrosion occurs when a metal is exposed

to hydrogen environment. Iron liberates atomic hydrogen with hydrogen sulphide in the following way.



Hydrogen diffuses into the metal matrix in this atomic form and gets collected in the voids present inside the metal. Further, diffusion of atomic hydrogen makes them combine with each other and forms hydrogen gas.



Collection of these gases in the voids develops very high pressure, causing cracking of metal.

3. Liquid metal corrosion: This is due to chemical action of flowing liquid metal at high temperatures on solid metal or alloy. Such corrosion occurs in devices used for nuclear power. The corrosion reaction involves either: (i) dissolution of a solid metal by a liquid metal or (ii) internal penetration of the liquid metal into the solid metal. Both these modes of corrosion cause weakening of the solid metal.

Electrochemical Corrosion (Wet Corrosion)-

This types of corrosion occurs (i) where a conducting liquid is in contact with metal or (ii) When two dissimilar metals or alloys are either immersed or dipped partially in a solution.

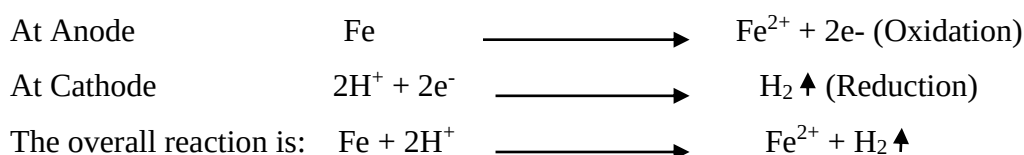
In this process a separate anode and cathode areas are form between which current flows through conducting solution. Corrosion always occurs at the anodic areas of metal through oxidation reaction and release electron. The liberated electron from anode, move towards cathode, where reduction take place. The overall theory is known as electrochemical theory of corrosion. It is also known as wet theory of corrosion because corrosion takes place in presence of moisture.

Electrochemical corrosion involves flow of free electrons from anode to cathode. At anode dissolution of metal as metallic ions with the liberation of free electrons.

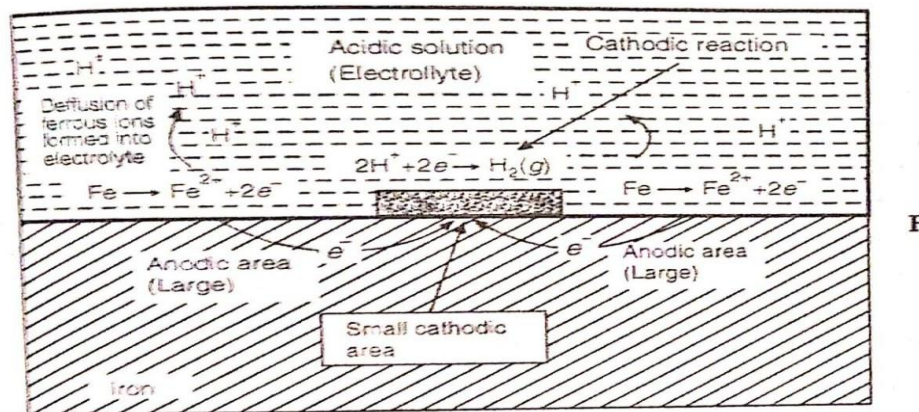


At cathode hydrogen liberation or absorption of oxygen takes place.

a) Hydrogen Liberation: In this type of corrosion, evolution of hydrogen occurs in acidic medium. Let's considering the metal Fe, anodic reaction is dissolution of iron as ferrous ions with liberation of electrons.

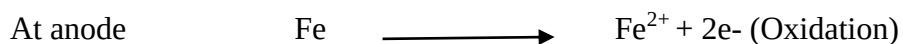


The electrons released flow through the metal from anode to cathode, whereas H^+ ions of acidic solution are eliminated as hydrogen gas.

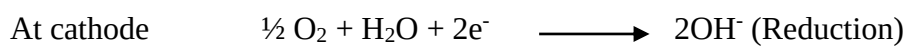


Mechanism of wet corrosion- by hydrogen evolution

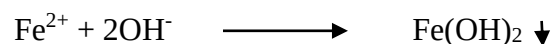
b) Absorption of Oxygen: The rusting of iron takes place in neutral aqueous solution of electrolytes (like NaCl solution) in presence of atmospheric oxygen. If the metal surface develops cracks, anodic areas are created on the surface whereas the rest metal parts act as cathodes.



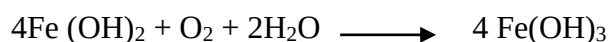
The released electrons flow from anode to cathode through iron metal.



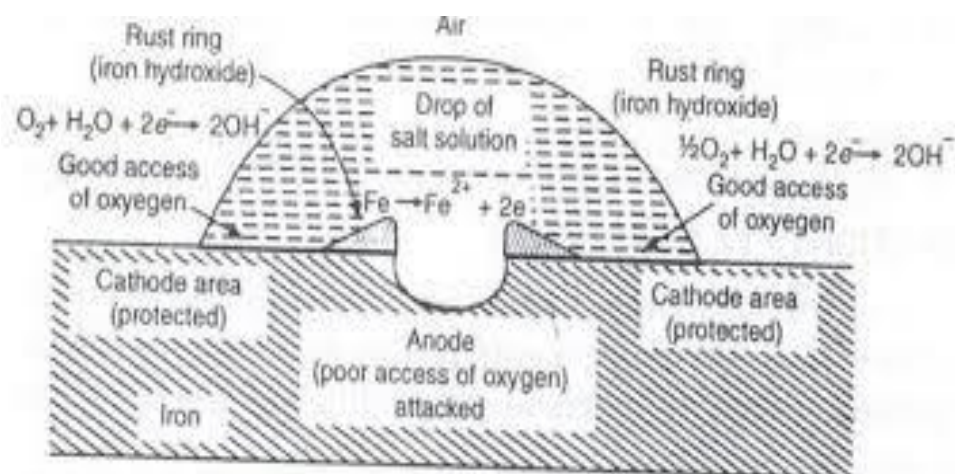
$Fe(OH)_2$ precipitate when Fe^{2+} and OH^- ions reacts with each other



If oxygen is in excess, ferrous hydroxide is easily oxidized to ferric hydroxide.



The product called yellow rust, actually corresponds to $Fe_2O_3 \cdot 3H_2O$



Mechanism of wet corrosion- by oxygen absorption

Difference between (dry) chemical and (wet) electrochemical corrosion

Sr. No.	Chemical Corrosion	Electrochemical Corrosion
1	It occurs in dry condition.	It occurs in the presence of moisture or electrolyte.
2	It is due to the direct chemical attack of the metal by the environment.	It is due to the formation of a large number of anodic and cathodic areas.
3	Even a homogeneous metal surface gets corroded.	Heterogeneous (bimetallic) surface alone gets corroded.
4	Corrosion products accumulate at the place of corrosion.	Corrosion occurs at the anode while the products are formed elsewhere.
5	It is a self controlled process.	It is a continuous process.
6	It adopts adsorption mechanism	It follows electrochemical reaction.
7	Formation of mild scale on iron surface is an example	Rusting of iron in moist atmosphere is an Example.

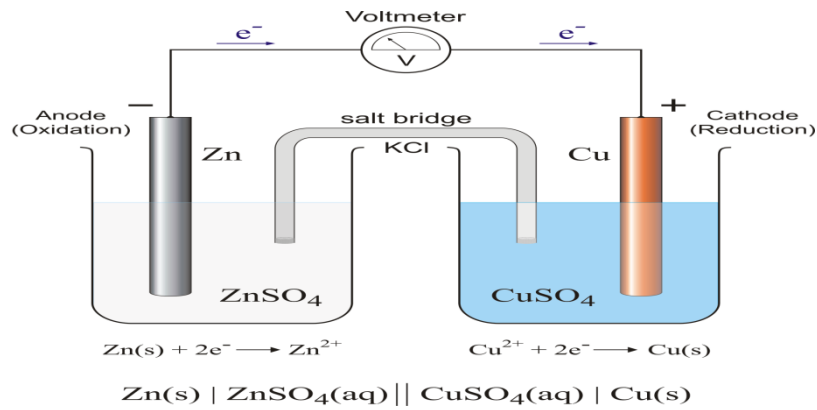
Galvanic Corrosion:

This corrosion occurs when two dissimilar metals are connected and dipped in an electrolytic solution. It is also known as bimetallic corrosion.

When two metals having different electrode potentials are in contact with each other in the presence of an electrolyte, one metal acts as a cathode whereas the other acts as an anode. This leads to an electrochemical reaction as a result of which, the anode is dissolved into the electrolyte. This electrochemical process is called galvanic corrosion of metals.

Mechanism of Galvanic Corrosion

When two metals come in contact in the presence of an electrolyte, the more reactive metal acts as an anode whereas the less reactive metal acts as a cathode. Galvanic Corrosion can be used beneficially to protect a cathodic metal from corrosion.



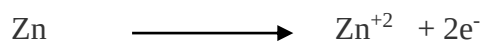
The metal higher in electrochemical series forms in anode and is corroded whereas the metal lower in galvanic series act as cathode and is protected.

Example- In case of Zn-Cu couple, Zn is corroded and Cu is protected.

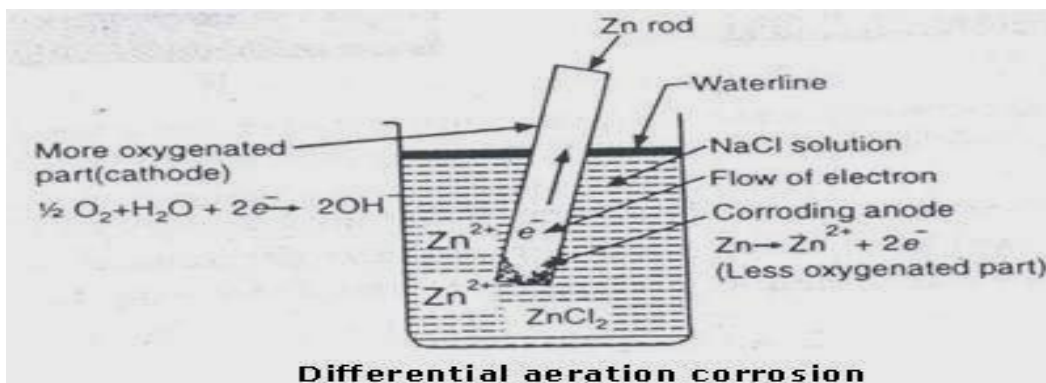
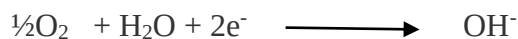
Concentration Cell Corrosion:

Concentration cell corrosion occurs when two or more areas of a metal surface are in contact with different **concentrations** of the same solution. It is due to electrochemical attack on the metal surface, exposed to an electrolyte of varying concentrations. It occurs when one part of metal is exposed to a different air concentration from the other part. This causes a difference in potential between differently aerated areas. It has been found experimentally that poor-oxygenated parts are anodic.

\Example- If Zinc metal is partially immersed in a dilute solution of NaCl and the solution is not stirred, the parts above and loosely adjacent to the water line are well aerated and act as cathode. Area deep inside the solution are anodic, since the concentration of oxygen is smaller. Thus a potential is developed which causes a flow of current between two axes of the same metal Zn dissolve at the anode area.

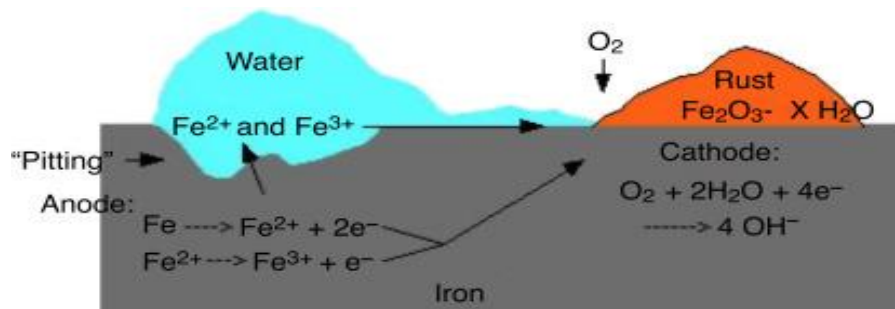


At the cathode area oxygen takes up electrons to form hydroxy ions-

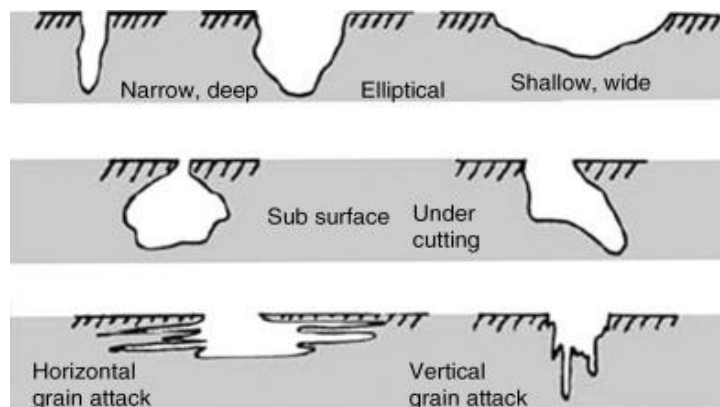


Pitting Corrosion

Pitting corrosion is a localized attack where the corrosion dissolves the metal, forming pits or holes. Pitting is usually the result of the breakdown or cracking of the protective film on a metal at specific points. This type of corrosion is dangerous because it is difficult to detect due to the corrosion products that may cover the pits, difficult to design against the formation of pits, as well as to predict when this type of corrosion may occur. The electrochemical oxidation–reduction process of pitting corrosion on surfaces of metals in an aqueous media is shown as-

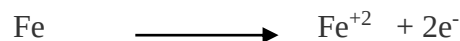


Pitting Corrosion



Types of Pits

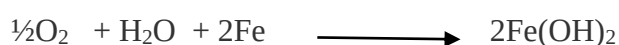
At Anode removal of electrons takes place



At the cathode area oxygen takes up electrons to form hydroxy ions-



Overall chemical reaction is-

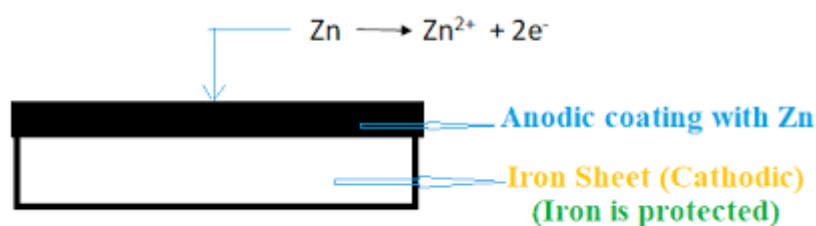


Corrosion Prevention

a) **Metal Coating** In order to protect base metal from corrosion metal coating process has been adopted. The coating of metal can be achieved by following processes

i) **Anodic metal coating (Galvanization):**

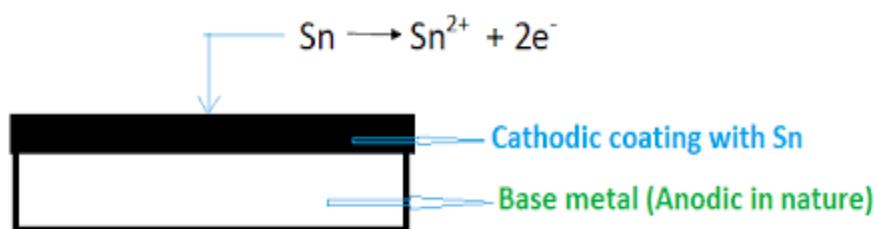
Metal coating which is anodic to base metal, known as anodic metal coating. In order to protect a metal from corrosion, any metal (anodic in nature) which is higher in the galvanic series like Zn is coated on the surface of metal (base metal/ cathodic in nature). The best example of anodic metal coating is Galvanization. Galvanization is a process in which coating of base metal (to be protected) takes place by dipping the base metal (cathodic in nature) into the molten zinc (anodic in nature).



Uses: - Roofing sheets, wires, pipes, bolts, screws, buckets, tubes etc. which are made of iron are protected from atmospheric corrosion by galvanizing them. But galvanized utensils cannot be used for preparing and storing food stuffs, especially acidic ones since zinc gets dissolved in all acids forming highly toxic compounds.

ii) **Cathodic metal coating (Tinning):**

Metal coating which is cathodic to base metal, known as cathodic metal coating. In order to protect a metal from corrosion, any metal (cathodic in nature) which is lower in the galvanic series like Sn, Cu, Ni etc, are coated on the surface of metal (base metal/ anodic in nature). The best example of cathodic metal coating is Tinning.



Tinning is a process in which coating of base metal (to be protected) takes place by dipping the base metal (anodic in nature) into the molten Tin (cathodic in nature).

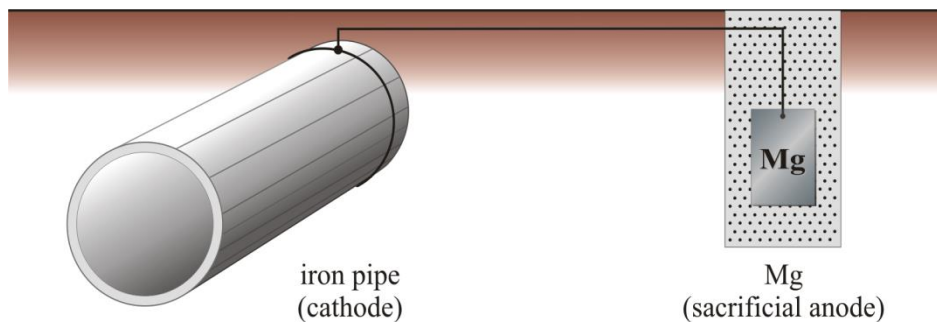
Uses:

- 1) Tin coated iron or steel articles or utensils are used for storing and preparing of food stuffs.

2) It is most widely used for protection of iron or steel from atmospheric corrosion in the form of roofing sheets, pipes, etc.

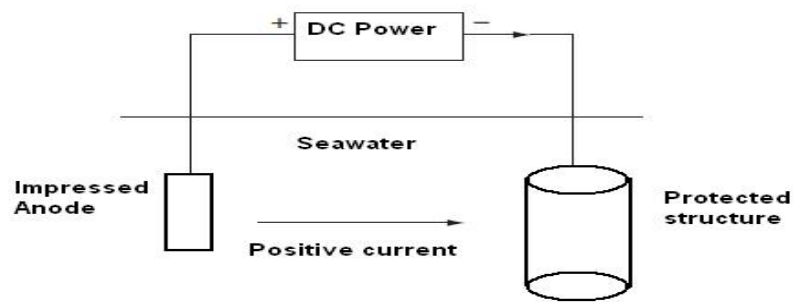
b) **Cathodic protection-** The principle involved in this method is to force the metal to be protected to behave like a cathode. There are two types of cathodic protections-

i) **Sacrificial anodic protection:** In this protection method, the metallic structure to be protected (base metal) is connected by a wire to a more anodic metal so that all the corrosion occurs at that anodic metal and metallic structure (cathode) is protected. The more active metal so employed is called sacrificial anode. The corroded sacrificial anode is replaced by a fresh one, when consumed completely. Metals commonly employed as sacrificial anode are Mg, Zn, Al and their alloys which possess low reduction potential (more reactive) and occupies higher end in electrochemical series. The underground water pipelines and water tanks are also protected by sacrificial anode method.



The anodic protection with sacrificial anode

ii) **Impressed current cathodic protection:** In this method, an impressed current from external source is applied in opposite direction to neutralize the corrosion current, and convert the corroding metal from anode to cathode. Thus the anodic corroding metal becomes cathodic and protected from corrosion. The impressed current is deriving from the external DC source. The negative terminal of DC source is connected with metal to be protected whereas the positive terminal of DC is connected to the anode. Here anode may be graphite, stainless steel, scrap iron etc. The impressed current protection method is used for water tanks, water & oil pipe lines, transmission line towers etc.



Impressed cathodic protection

Modification in Designs:

- The contact of two dissimilar metals in the presence of a corroding solution should be avoided. Thereby differential metal corrosion can be controlled.
- The anodic materials should have larger area compared to the cathode metal.
- The bolts, screws and nuts should be made from more noble metal.
- When two dissimilar metals in contact have to be used, they should be as close as possible to each other in the electrochemical series, so that the differential type corrosion can be minimized.
- The anodic metal should not be painted or coated when in contact with a dissimilar cathodic metal because any break in coating would lead to rapid localized corrosion.
- The metal part, as far as possible, should be free from tensile stresses to avoid stress corrosion.